described in section 10.3) and then move the cluster's centroid to the center of mass computed.

4. Repeat steps *classification* and *recentering* until no point, at step 2, switches to a different cluster, or the maximum number of iterations is reached.

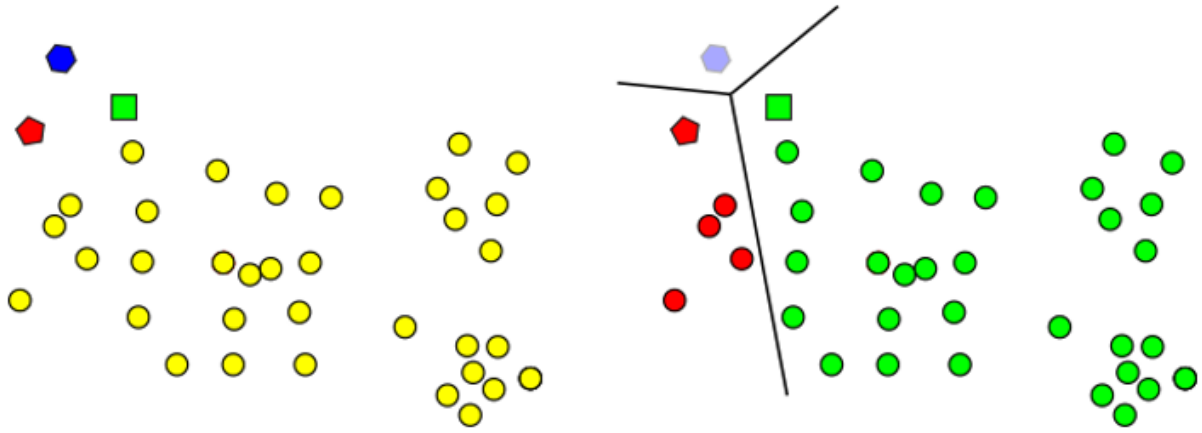


Figure 12.4.A very unlucky choice of the initial centroids (shown as polygons). All centroids are gathered together in one corner of the dataset. The black line shows an approximation of the border between the regions determined by each centroid (on each line, ideally, the distance between the centroids it separates is the same). Since the centroid represented with a blue hexagon is further away than the other two, no point will be assigned to it and so it will be removed from the list of centroids. The choice appears also unbalanced for the other two centroids, although the next update steps will (slowly) re-balance the situation, moving the green square towards the center of the cluster on the right.

Steps 2-4 of this algorithm are a deterministic heuristic that computes exact distances between points and centroids, and